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# Mini AI Project - CRISP-DM

**Specialization: Data Science AI**

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# Predictive Analysis of Customer Churn in the Banking Sector: A Machine Learning Approach

Academic Year: 2025-2026

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# General Introduction

In recent years, the rapid evolution of information technologies and the massive growth of data have profoundly transformed decision-making processes across various sectors. In particular, the banking sector generates large volumes of customer data through daily transactions, interactions, and digital services. When properly analyzed, this data represents a valuable asset that can support strategic decisions and improve business performance.

One of the major challenges faced by banks today is customer retention. In a highly competitive environment, customers can easily switch from one financial institution to another, leading to customer churn. This phenomenon has a direct impact on profitability, operational stability, and long-term growth. As a result, anticipating customer churn and understanding its underlying causes have become essential objectives for modern banks.

Machine learning and data science techniques provide powerful tools for extracting meaningful patterns from historical data and predicting future customer behavior. By leveraging these techniques, banks can move from reactive strategies to proactive decision-making, enabling targeted retention actions and improved customer satisfaction.

This project focuses on the prediction of customer churn in the banking sector using a data-driven approach. The work follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology, which offers a structured and systematic framework for conducting data science projects. This methodology emphasizes business understanding, data analysis, modeling, evaluation, and deployment, ensuring that technical solutions remain aligned with real business needs.

The objective of this report is to analyze customer data,identify key factors influencing churn, and build predictive models capable of estimating the likelihood of a customer leaving the bank. Through this approach, the project aims to demonstrate how machine learning can support decision-making and contribute to improving customer retention strategies in the banking domain.

The remainder of this report is organized according to the different phases of the CRISP-DM methodology, providing a clear and logical progression from problem definition to model evaluation and interpretation of results.

# Chapter 1

# Problematic and Business Understanding

## 1.1 Introduction

The banking sector is currently undergoing a massive digital transformation, leading to increased competition and unprecedented mobility for customers. With the proliferation of digital banking options, clients can switch financial institutions with minimal friction. Consequently, customer retention has evolved from a secondary operational concern to a primary strategic imperative. Acquiring a new customer is widely recognized as being significantly more expensive than retaining an existing one. To maintain profitability and long-term sustainability, financial institutions must shift from reactive customer service models to proactive retention strategies. This chapter outlines the business context of customer churn, introduces the chosen methodology for this project, defines the core problem, and establishes the objectives and hypotheses that will guide our analytical approach.

## 1.2 CRISP-DM Methodology

To ensure a structured, systematic, and industry-standard approach to this problem, this project adopts the **CRISP-DM (Cross-Industry Standard Process for Data Mining)** methodology. CRISP-DM is a robust framework designed to guide data science projects from the initial business problem to the deployment of the solution.

The methodology is cyclical and consists of six major phases:

- **Business Understanding:** Defining the project objectives and requirements from a business perspective, and translating this knowledge into a data mining problem definition.
- **Data Understanding:** Collecting initial data, evaluating its quality, and discovering initial insights.
- **Data Preparation:** Constructing the final dataset from the raw data through cleaning, transformation, and feature engineering.
- **Modeling:** Selecting and applying various modeling techniques, and calibrating their parameters to optimal values.

- **Evaluation:** Evaluating the model to ensure it properly achieves the business objectives set in the first phase.
- **Deployment:** Integrating the finalized model into the business workflow for practical use.

This project begins with a deep dive into the first phase: **Business Understanding**.

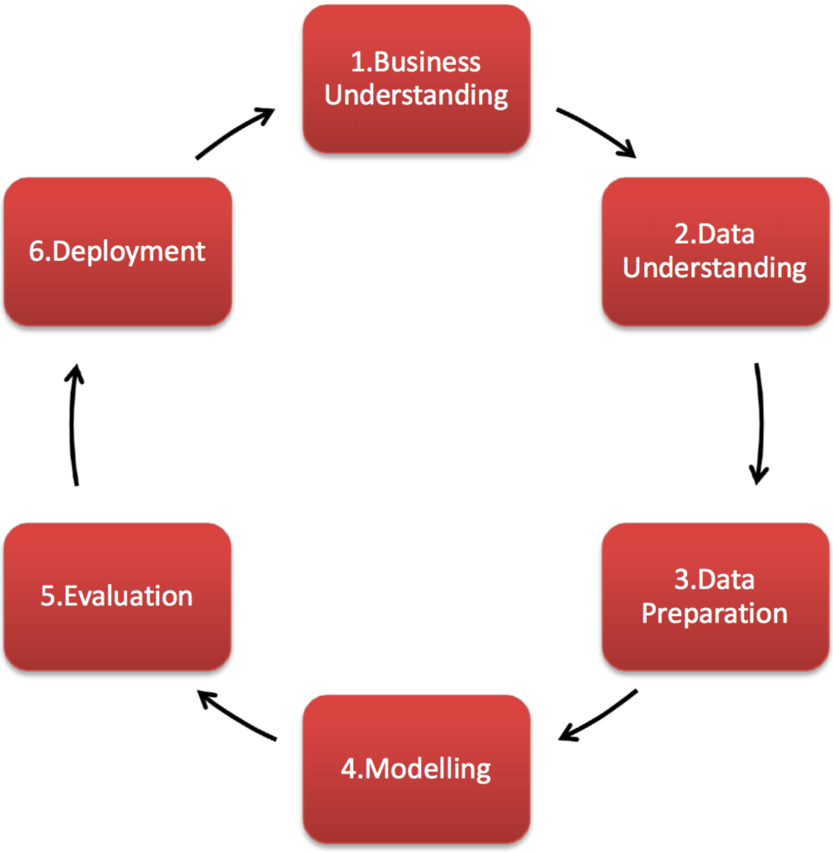


Figure 1.1: CRISP-DM Process Model

## 1.3 Context and Motivation

In the banking sector, customer retention has become a major strategic challenge due to increasing competition, digital transformation, and the ease with which customers can switch between financial institutions. Acquiring new customers generally requires higher costs compared to retaining existing ones. Therefore, banks are increasingly focused on understanding customer behavior in order to anticipate dissatisfaction and reduce customer churn.

Customer churn refers to the phenomenon where clients stop using the bank's services or close their accounts. High churn rates negatively impact the bank's revenue, profitability, and long-term sustainability. Consequently, the ability to identify customers who are at risk of leaving the bank represents a critical business issue for financial institutions.

With the availability of large volumes of customer data, data science and machine learning techniques offer an opportunity to support decision-making by transforming historical data into actionable insights. This project is carried out within the finance/banking domain and follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology, which provides a structured and systematic approach to solving data-driven problems.

## 1.4 Business Problem Definition

In a highly competitive banking environment, customer retention has become a critical challenge for financial institutions. Losing customers not only results in direct revenue loss but also increases acquisition costs, making churn prevention a strategic priority for banks.

From a business perspective, banks face several key questions:

- Which customers are most likely to leave the bank?
- What are the main factors that contribute to customer churn?
- How can the bank take proactive actions to retain high-risk customers?

Currently, many banks rely on reactive retention strategies based on manual analysis or generalized marketing campaigns. These approaches are often inefficient, costly, and poorly targeted, as they do not consider individual customer behavior and risk profiles.

A predictive approach based on data analysis and machine learning techniques can help overcome these limitations. By anticipating customer churn before it occurs, banks can focus their retention efforts on customers who are most likely to leave, optimize resource allocation, and design personalized intervention strategies.

Therefore, the core business problem addressed in this project is the anticipation of customer churn in the banking sector in order to support strategic decision-making, improve customer loyalty, and enhance overall business performance.

## 1.5 Existing Solutions vs. Proposed Solution

### 1.5.1 Existing Solutions

Traditionally, banks have relied on manual analysis, generalized marketing campaigns, or basic heuristic rules (such as RFM analysis: Recency, Frequency, Monetary value) to identify at-risk customers. These approaches are often:

- **Reactive:** Intervening only after a customer has expressed dissatisfaction or initiated the account closure process.
- **Costly and Inefficient:** Applying blanket retention strategies (such as generic mass emails or promotional offers) that are poorly targeted and lead to inefficient resource allocation.
- **Limited in Scope:** Unable to capture complex, non-linear relationships in customer behavior and profiles.

### 1.5.2 Our Proposed Solution

To overcome the limitations of traditional methods, this project proposes a predictive approach leveraging **Machine Learning** and **Deep Learning**. By training advanced algorithms on historical customer data, the proposed solution aims to:

- **Predict Churn Proactively:** Identify the probability of a customer leaving before churn actually occurs.
- **Capture Complex Patterns:** Utilize Deep Learning architectures to uncover hidden patterns within demographic, financial, and behavioral data that traditional rule-based systems fail to detect.
- **Enable Targeted Interventions:** Allow the bank to segment high-risk customers and allocate retention budgets efficiently through highly personalized strategies.

## 1.6 Problematic Statement

In accordance with the CRISP-DM methodology, the business problem addressed in this project must be clearly defined, realistic, and oriented toward decision-making. The objective is to translate a concrete business need into an analytical problem that can be solved using data-driven approaches.

**Problematic:**

*Can customer churn in a bank be accurately predicted using historical customer data in order to identify clients at risk of leaving and enable proactive and targeted retention strategies?*

This problematic focuses on anticipating customer behavior rather than reacting after churn has already occurred. By identifying high-risk customers in advance, the bank can optimize its retention efforts, reduce revenue loss, and improve customer loyalty.

In this context, the target variable is the **customer churn status**, which indicates whether a customer has left the bank or remains active. This variable is typically represented as a binary outcome and serves as the main output of the predictive model.

## 1.7 Business Objectives

The primary objective of this project is to address the issue of customer churn in the banking sector by leveraging historical customer data to support informed business decisions.

More specifically, the business objectives of this project are as follows:

- To analyze and understand the key factors that influence customer churn in the banking sector.
- To predict whether a customer is likely to leave the bank using data-driven models.
- To enable the bank to prioritize and target retention actions toward customers identified as high risk.
- To support strategic decision-making through actionable insights derived from data analysis.

These objectives are fully aligned with the **Business Understanding** phase of the CRISP-DM methodology. They emphasize business value creation, customer retention, and decision support rather than focusing solely on technical or algorithmic implementation.

## 1.8 Business Hypotheses

Based on domain knowledge and common practices in the banking industry, several business hypotheses are formulated to guide the analysis and modeling process. These hypotheses reflect assumptions about customer behavior and their potential relationship with customer churn.

- **Customer Activity Level Influences Churn:** Customers with low transaction activity are more likely to leave the bank compared to highly active customers.
- **Customer Tenure Affects Loyalty:** Customers who have been with the bank for a shorter period are more likely to churn than long-term customers, as loyalty typically increases over time.
- **Product Engagement Impacts Retention:** Customers who use fewer banking products (such as credit cards, savings accounts, or loans) have a higher probability of leaving the bank.

These hypotheses will be evaluated, validated, and refined throughout the subsequent phases of the CRISP-DM process, particularly during the data understanding, modeling, and evaluation stages.

## 1.9 Required Data

To effectively address the problematic, the following types of data would ideally be required:

- Customer demographic information (age, gender, location)
- Account information (balance, account type, tenure)
- Transaction and activity history
- Product usage data
- Customer churn label (active or churned)

These data would typically be produced by the bank's information systems and updated on a regular basis.

## 1.10 Conclusion

This chapter established the business context, motivation, and problematic of the project within the banking domain. By defining the problem through the lens of the CRISP-DM methodology, comparing existing methods to our proposed Machine Learning and Deep Learning solution, and outlining clear business objectives, a solid foundation has been laid. The next phase of the project, detailed in the following chapter, will focus on acquiring the specific dataset, understanding its attributes, and exploring the data to validate our initial hypotheses..

# Chapter 2

# Data Understanding & Preparation

## 2.1 Introduction

Data Understanding is the second stage of CRISP-DM methodology. Once the business goals are identified, and now is to gather, describe, investigate, and check the quality of data. The aim of this step is to get acquainted with the dataset, find preliminary insights, confirm the business assumptions made in the last chapter, and identify any possible data quality problems. In this chapter, the features are described. of the banking data and introduces a comprehensive Exploratory Data Analysis (EDA) to know the drivers of customer churn.

## 2.2 Dataset Description

For this project, a dataset obtained from Kaggle is utilized as an approximation of a real-world banking churn problem.[1] While it may not capture all operational complexities of an actual bank, it provides relevant features commonly used in churn analysis and allows for meaningful experimentation with predictive models. The main raw attributes of this dataset are :

- **RowNumber**: Row Numbers from 1 to 10000.
- **CustomerId**: Unique Ids for bank customer identification.
- **Surname**: Customer's last name.
- **CreditScore**: Credit score of the customer.
- **Geography**: The country from which the customer belongs.
- **Gender**: Male or Female.
- **Age**: Age of the customer.
- **Tenure**: Number of years for which the customer has been with the bank.
- **Balance**: Bank balance of the customer.
- **NumOfProducts**: Number of bank products the customer is utilising.

[1] https://www.kaggle.com/datasets/shrutimechlearn/churn-modelling

- **HasCrCard**: Binary Flag for whether the customer holds a credit card with the bank or not.
- **IsActiveMember**: Binary Flag for whether the customer is an active member with the bank or not.
- **EstimatedSalary**: Estimated salary of the customer in Dollars.
- **Exited**: Binary flag 1 if the customer closed account with bank and 0 if the customer is retained.

The dataset is suitable for this study for several reasons:

- It represents realistic customer profiles, including demographic, financial, and account-related information.
- It contains a clearly defined **churn variable**, which serves as the target for predictive modeling.
- It enables experimentation with data-driven decision-making and evaluation of predictive approaches for customer retention.

However, it is important to acknowledge certain limitations of the dataset:

- Customer behavior is simplified and may not fully capture the complexity of real-world banking operations.
- Some relevant factors influencing churn in practice may be absent.

These limitations are taken into consideration during data analysis and modeling to ensure realistic interpretation of the results.

## 2.3 Dataset Overview and Quality Assessment

The dataset used for this project consists of **10,000 customer records** and **14 attributes**, encompassing demographic details, account information, and the target variable *Exited*.

Initial data profiling revealed the following statistics and quality checks:

- **Completeness:** A check for missing values across all columns showed exactly **0 null values**. The dataset is complete and requires no imputation for missing data.
- **Descriptive Statistics:**
  - The average customer age is approximately **39 years**, ranging from 18 to 92.
  - The average account balance is around **76,485**, although 25% of customers have a balance equal to 0.
  - About **51.5%** of customers are classified as active members.
  - The target variable (*Exited*) has a mean of **0.2037**, indicating an overall churn rate of approximately **20.4%**.

## 2.4 Exploratory Data Analysis (EDA)

To uncover underlying patterns and understand how different features relate to customer churn, we conducted a comprehensive Exploratory Data Analysis.

### 2.4.1 Target Variable Distribution

The primary focus of our analysis is the *Exited* column, which indicates whether a customer has churned (1) or been retained (0).

- **Observation:** The visualization of the target column highlights a significant class imbalance. Approximately 80% of customers were retained, while only 20% churned. Because accuracy is not a reliable metric for imbalanced datasets, this observation confirms the necessity of applying data balancing techniques (such as SMOTE) during the modeling phase.

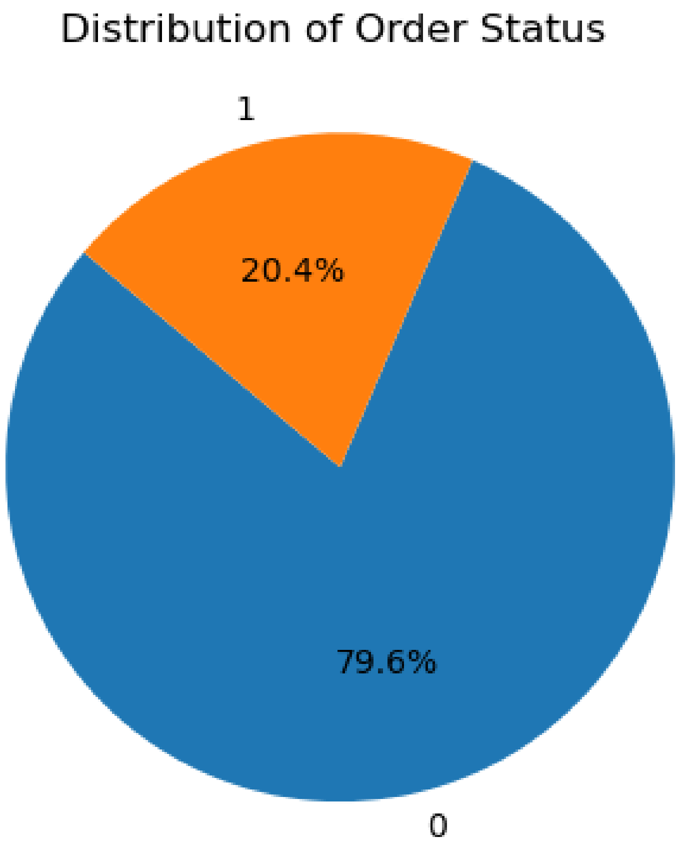


Figure 2.1: Distribution of the Target Variable (Exited)

### 2.4.2 Outlier Detection

We analyzed the numerical features using box plots to detect the presence of severe anomalies.

- **Observation :** There are no significant outliers across most numerical features. The *Age* variable does display some statistical outliers (older customers), but these represent natural demographic variations rather than data errors. Therefore, no corrective action is necessary.

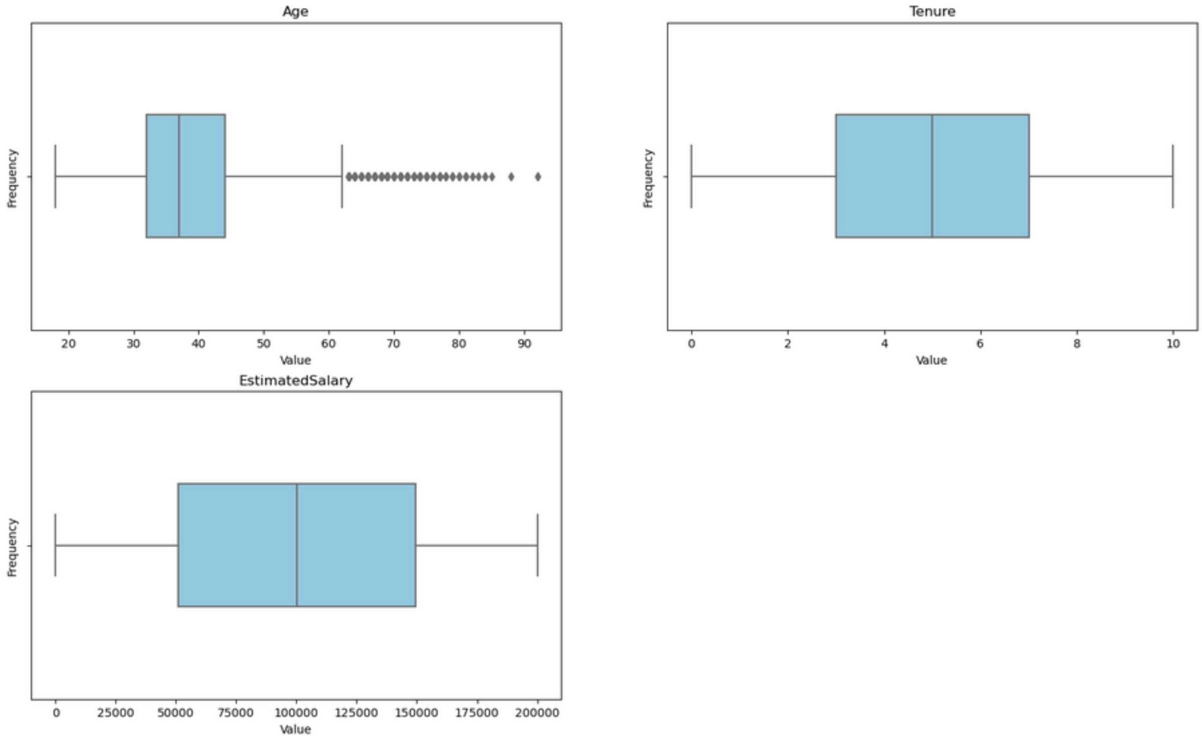


Figure 2.2: Boxplot Visualization for Outlier Detection

### 2.4.3 Feature Distribution by Churn Status

To understand what drives a customer to leave, we compared the distributions of various features between churned and retained customers.

- **Credit Score :** The credit score distribution between customers who churned and those who did not is highly similar. There is no stark contrast between the two groups, implying that credit score alone is not a strong predictor of churn. However, categorical grouping shows that a majority of customers who churned fall into the *High* credit score group.
- **Tenure :** The tenure distribution suggests that customers who have been with the bank for a shorter period are slightly more likely to churn than long-term customers.
- **Age:** Age demonstrates a pronounced difference. Older customers appear significantly more likely to churn compared to younger customers.
- **Balance:** The balance distribution is notably different. Churned customers tend to have higher account balances on average. This suggests that customers with higher financial stakes may still be at risk of leaving.
- **Estimated Salary:** The estimated salary distributions do not show a significant difference between the two groups, suggesting that income level is not a primary factor in customer churn.

- **Tenure vs. Balance :** A scatter plot analyzing the interaction between tenure and balance indicates no clear linear pattern or correlation. These variables do not strongly predict churn when combined in a simple linear manner.

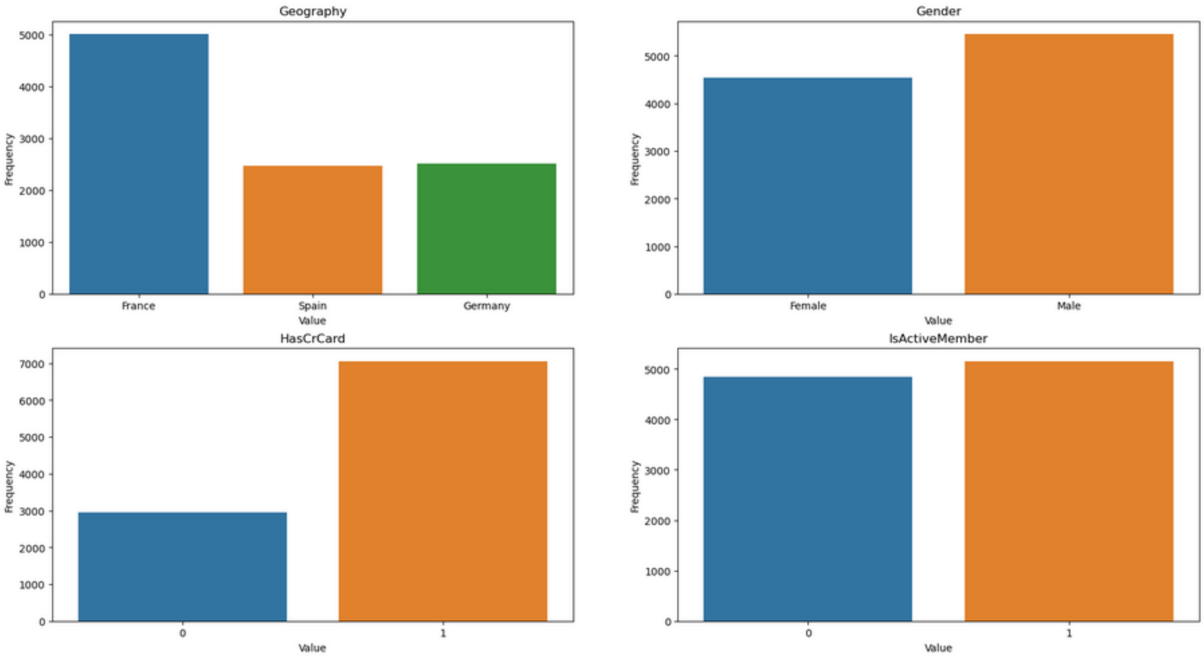


Figure 2.3: Feature Distributions by Churn Status

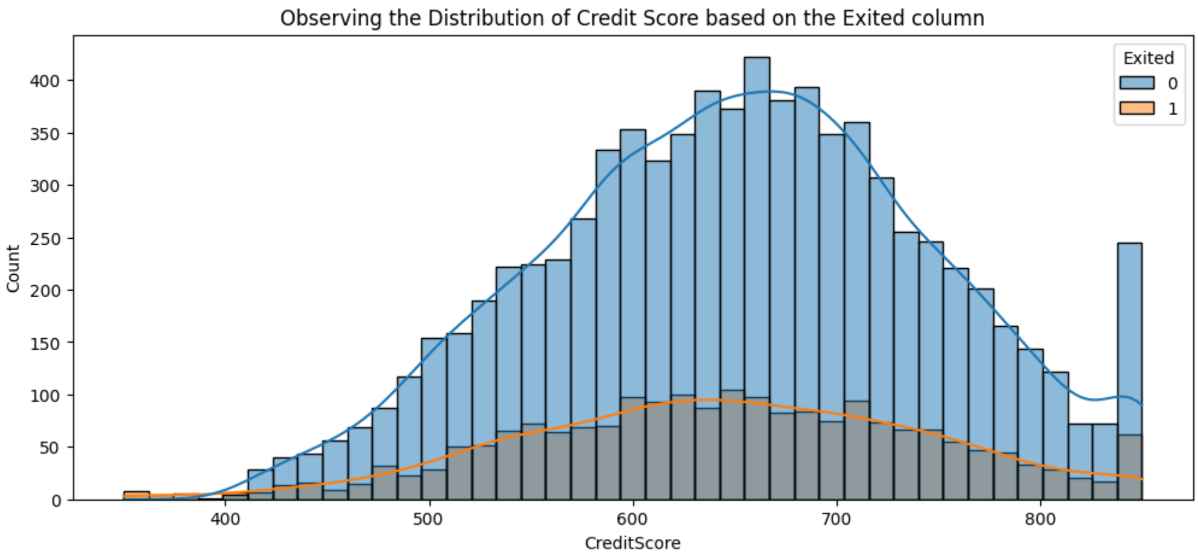


Figure 2.4: Credit Score Distribution by Churn

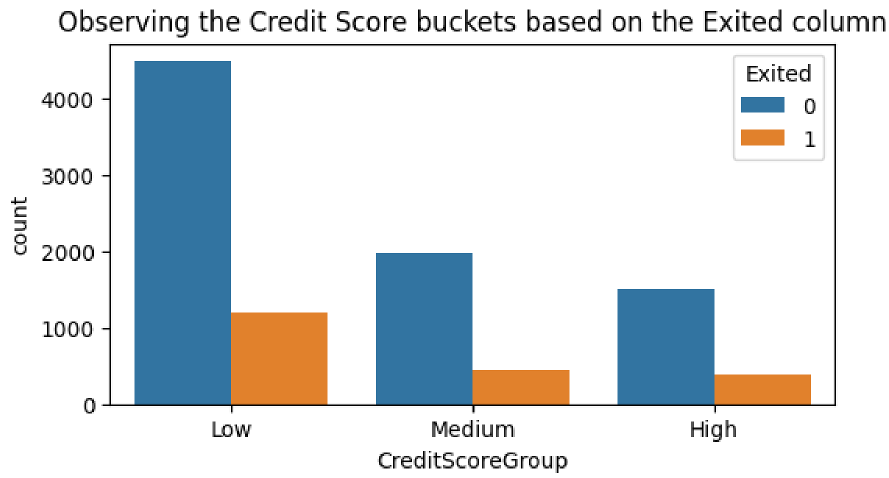


Figure 2.5: Credit Score Category Distribution by Churn

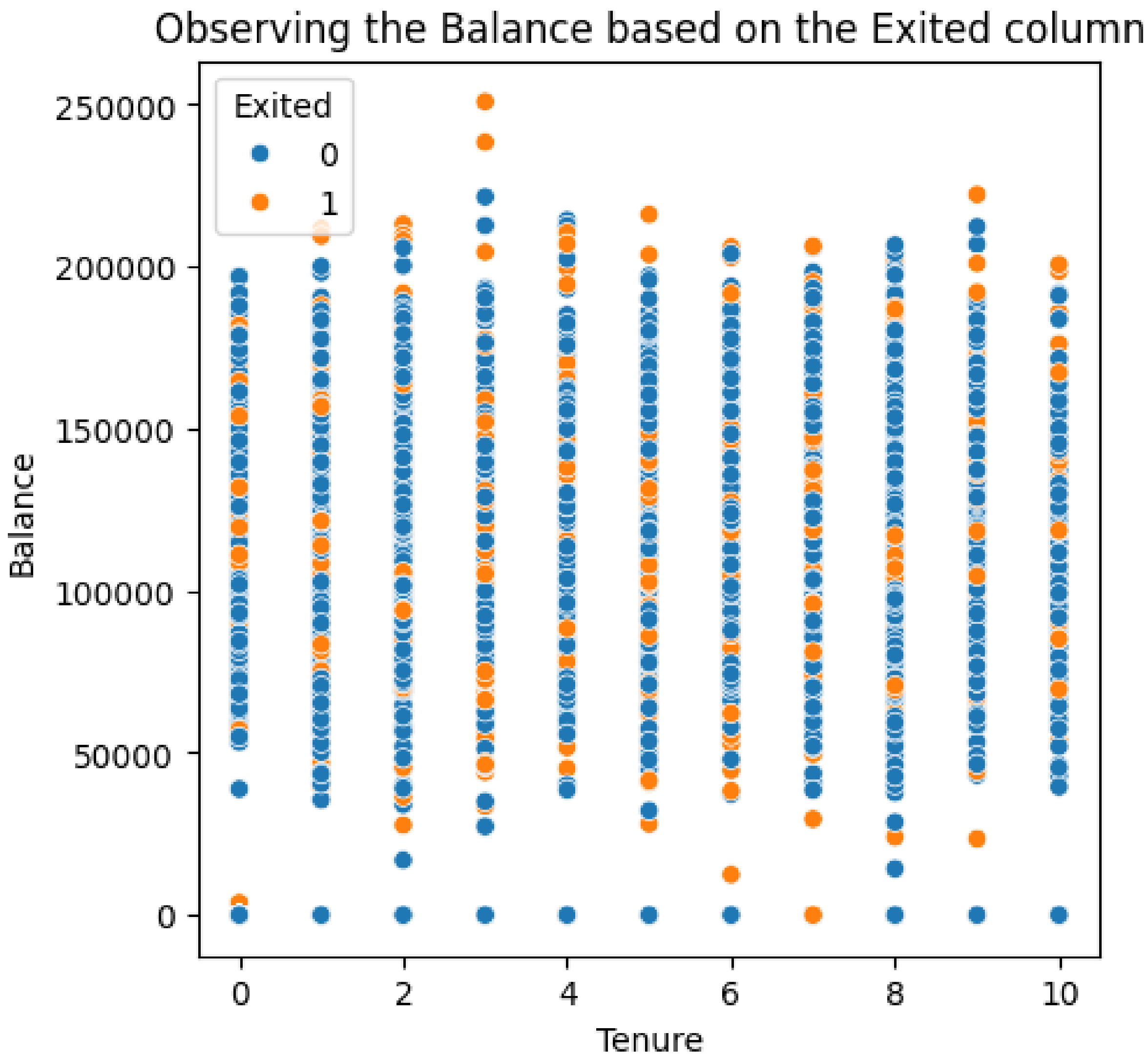


Figure 2.6: Scatter Plot: Tenure vs Balance

sectionFeature Engineering

To extract deeper insights and provide our predictive models with more meaningful variables, we engineered several new composite features:

- **Credit Utilization:** The ratio of balance to credit score, providing insight into how much of the available credit the customer is using.
- **Interaction Score:** A composite score based on the number of products, active membership status, and credit card possession. This variable captures overall customer engagement, where higher engagement is hypothesized to reduce churn risk.
- **Balance to Salary Ratio:** The ratio of the customer's balance to their estimated salary, indicating how significant the customer's holdings are relative to their income.
- **Credit Score–Age Interaction:** An interaction term between credit score and age to explore whether the influence of credit score on churn varies across different age groups.

## 2.5 Correlation Analysis

We analyzed the linear relationships between all numerical variables, including the engineered features, using a correlation matrix heatmap.

- **Observations :**
  - **Age**

    Older customers show a moderate positive correlation with churn (0.28).

    Possible causes:
    * Older customers may change banks after retirement.
    * They may have different financial needs.
    * Reduced loyalty to digital banking services.
  - **CreditScoreAgeInteraction**

    This engineered feature shows a positive correlation with churn (0.23), making it one of the strongest predictive engineered features.

    Possible causes:
    * The impact of credit score may depend on customer age.
    * Younger customers with low credit scores may not churn as much as older customers with similar scores.
    * Credit risk perception varies across age groups.
  - **CreditUtilization**

    Shows a weak to moderate positive correlation (0.12).

    Possible causes:
    * Customers using a high percentage of their credit may experience financial stress.
    * High credit usage may indicate dependency on credit facilities.

    * Financial pressure may motivate customers to search for better banking conditions.

- **Balance**
  Shows a weak positive correlation with churn (0.11).
  Possible causes:
    * Customers with high balances may switch banks to seek better interest rates.
    * Wealthier customers often have more banking alternatives.
    * High balance alone does not guarantee customer loyalty.
- **BalanceToSalaryRatio**
  Shows a very weak positive correlation (0.02).
  Possible causes:
    * Customer income estimation may not be perfectly accurate.
    * Financial behavior is complex and cannot be explained by simple ratios alone.
    * The relationship with churn may be non-linear.
- **Tenure**
  Shows very weak negative correlation (-0.014).
  Possible causes:
    * Customers who stay longer tend to be slightly more loyal.
    * However, tenure alone is not a strong churn predictor.
- **CreditScore**
  Shows weak negative correlation (-0.027).
  Possible causes:
    * Customers with higher credit scores are slightly more financially stable.
    * They are marginally less likely to churn.
- **NumOfProducts**
  Shows weak negative correlation (-0.047).
  Possible causes:
    * Customers using more products are generally more engaged.
    * Product diversification increases customer dependency on the bank.
- **InteractionScore**
  Shows a negative correlation (-0.12).
  Possible causes:
    * Higher engagement with bank services reduces churn probability.
    * Customers with multiple products and active usage tend to be more loyal.
- **IsActiveMember**
  Shows one of the strongest negative correlations (-0.15).
  Possible causes:
    * Active customers use banking services regularly.

* Inactive customers are more likely to leave.
* Customer engagement level is a strong loyalty indicator.

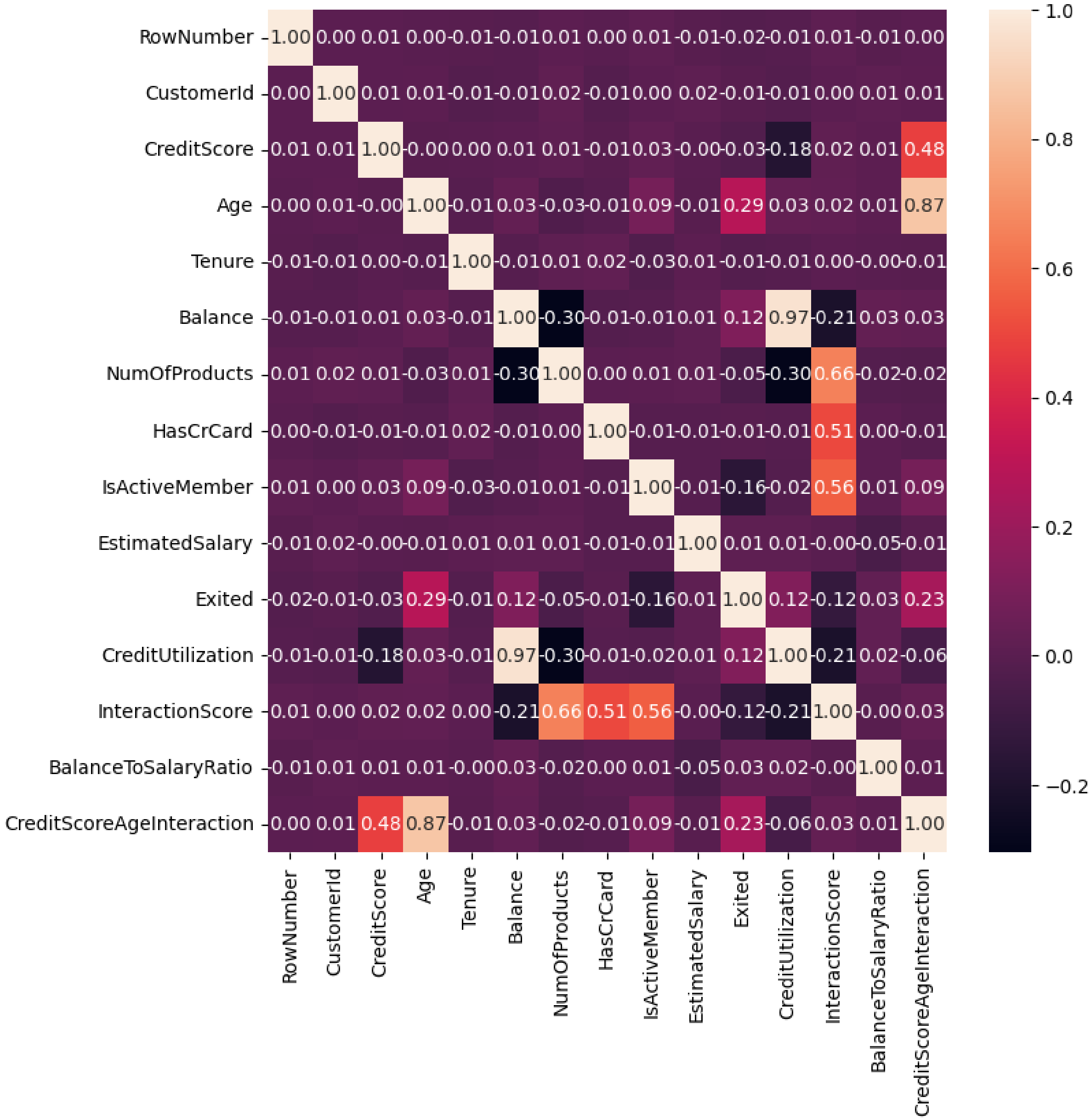


Figure 2.7: Correlation Matrix Heatmap

## 2.6 Conclusion

The Data Understanding phase has provided critical insights into customer behavior. While variables such as estimated salary and standalone credit scores do not strongly determine churn, features like **Age**, **Account Balance**, and **Active Membership status** play a significant role.

# Chapter 3

# Modeling

## 3.1 Introduction

The fourth phase of the CRISP-DM methodology is **Modeling**. Building on the insights gained during data understanding and the engineered features prepared in the previous phase, this chapter focuses on selecting, designing, and training predictive models capable of identifying customers at risk of churning.

This chapter presents two complementary families of models:

- **Classical Machine Learning models** with systematic hyperparameter optimisation via GridSearchCV.
- **Artificial Neural Network (ANN) architectures** of increasing complexity, trained with early stopping and adaptive learning rate scheduling.

Six models in total are trained and prepared for evaluation. The complete pipeline from data preprocessing to model persistence is described step by step, following CRISP-DM's emphasis on reproducibility and traceability.

## 3.2 Data Preprocessing Pipeline

Before any model is trained, a consistent preprocessing pipeline is applied to guarantee data quality and comparability across all experiments.

### 3.2.1 Feature Encoding

The two categorical variables present in the dataset are encoded as follows:

- **Geography** (France, Germany, Spain) is encoded using `LabelEncoder`, yielding integer codes 0, 1, and 2 respectively.
- **Gender** (Female, Male) is encoded as 0 and 1 respectively.

The three identifier columns *RowNumber*, *CustomerId*, and *Surname* are dropped, as they carry no predictive signal.

### 3.2.2 Train/Test Split

The dataset is divided into a training set (80%) and a held-out test set (20%) using a **stratified split** (`stratify=y`, `random_state=42`). Stratification ensures that the 20.4% churn rate is preserved in both subsets, preventing evaluation bias on the minority class.

### 3.2.3 Feature Scaling

All features are standardised using `StandardScaler` (zero mean, unit variance). The scaler is fitted exclusively on the training set and then applied to transform both the training and test sets, avoiding data leakage.

### 3.2.4 Handling Class Imbalance with SMOTE

As established in Chapter 2, the target variable is heavily imbalanced: approximately 80% of customers belong to the retained class and only 20% to the churned class. Training a model on such a distribution would lead it to trivially predict the majority class, resulting in deceptively high accuracy but poor detection of actual churners.

To address this, **SMOTE** (Synthetic Minority Over-sampling Technique) is applied exclusively to the training set after scaling. SMOTE creates synthetic samples for the minority class (churned customers) by interpolating between existing minority-class observations in feature space.

- **Before SMOTE:** 6,370 retained — 1,630 churned
- **After SMOTE:** 6,370 retained — 6,370 churned (balanced)

**Important:** SMOTE is applied *only* on the training data. The test set remains untouched to reflect real-world class distributions during evaluation.

## 3.3 Classical Machine Learning Models

Three classical ML algorithms are selected for their complementary strengths in tabular classification tasks. Each model is tuned using **GridSearchCV** with **5-fold Stratified Cross-Validation** on the SMOTE-balanced training data, optimising the **F1-score** as the primary scoring criterion — a deliberate choice given the class imbalance context.

### 3.3.1 Model 1 — Random Forest

#### Algorithm Overview

Random Forest is an ensemble learning method based on **bagging** (Bootstrap AGGregatING). It builds a large number of decision trees, each trained on a random bootstrap sample of the training data and using a random subset of features at each split. The final prediction is determined by majority vote across all trees. This randomisation makes Random Forest robust to overfitting and capable of capturing complex non-linear relationships.

#### Hyperparameter Tuning

The following grid was explored:

Table 3.1: Hyperparameter Grid — Random Forest

| **Hyperparameter** | **Values Tested** | **Description** |
|---|---|---|
| `n_estimators` | 100, 200, 300 | Number of decision trees in the forest |
| `max_depth` | None, 10, 20 | Maximum depth of each tree |
| `min_samples_split` | 2, 5, 10 | Minimum samples required to split a node |
| `class_weight` | balanced | Adjusts weights inversely to class frequencies |

GridSearchCV evaluates all combinations using 5-fold stratified cross-validation and selects the configuration that maximises the mean F1-score across folds.

### 3.3.2 Model 2 — Gradient Boosting

#### Algorithm Overview

Gradient Boosting is an ensemble method based on **boosting**: trees are built *sequentially*, where each new tree is fitted to the residual errors of its predecessors. The final prediction is the weighted sum of all individual trees. Unlike Random Forest, which parallelises independent trees, Gradient Boosting refines predictions iteratively, making it particularly powerful on structured tabular data.

#### Hyperparameter Tuning

Table 3.2: Hyperparameter Grid — Gradient Boosting

| **Hyperparameter** | **Values Tested** | **Description** |
|---|---|---|
| `n_estimators` | 100, 200 | Number of boosting rounds |
| `learning_rate` | 0.05, 0.1, 0.2 | Shrinkage factor applied to each tree's contribution |
| `max_depth` | 3, 5, 7 | Maximum depth of each boosting tree |
| `subsample` | 0.8, 1.0 | Fraction of training samples used per tree |

### 3.3.3 Model 3 — Support Vector Machine (SVM)

#### Algorithm Overview

A Support Vector Machine finds the **optimal separating hyperplane** that maximises the margin between the two classes. Using the **Radial Basis Function (RBF) kernel**, SVM implicitly maps the input features into a higher-dimensional space, enabling it to capture non-linear decision boundaries that are not accessible with linear classifiers. The `probability=True` flag is set to enable probability calibration via Platt scaling, which is required for ROC-AUC computation.

### Hyperparameter Tuning

Table 3.3: Hyperparameter Grid — SVM (RBF Kernel)

| Hyperparameter | Values Tested | Description |
|---|---|---|
| `C` | 0.1, 1, 10, 100 | Regularisation: high C = tighter fit to training data |
| `gamma` | scale, auto, 0.01, 0.001 | RBF kernel width: high gamma = more localised boundary |
| `class_weight` | balanced | Up-weights minority class in the optimisation objective |

## 3.4 Artificial Neural Network Architectures

Three ANN architectures are designed and trained, progressing from a minimal baseline to a fully regularised deep network. This graduated approach allows direct observation of how architectural complexity, dropout regularisation, and batch normalisation each influence learning dynamics and generalisation performance.

All three ANNs share the following training configuration:

Table 3.4: Shared ANN Training Configuration

| Setting | Value | Justification |
|---|---|---|
| Optimiser | Adam (lr=0.001) | Adaptive learning rate, widely effective for deep learning |
| Loss function | Binary cross-entropy | Standard loss for binary classification |
| Batch size | 64 | Balances gradient accuracy and training speed |
| Maximum epochs | 100 | Upper bound; early stopping prevents overtraining |
| EarlyStopping | patience=10 | Halts training when val_loss stops improving |
| ReduceLROnPlateau | factor=0.5, patience=5 | Halves learning rate upon learning plateau |
| Output activation | Sigmoid | Outputs churn probability in [0, 1] |

A validation split of 15% is taken from the SMOTE-balanced training data for monitoring training progress and triggering callbacks.

### 3.4.1 ANN 1 — Shallow Network (Baseline)

#### Architecture

Table 3.5: ANN 1 Architecture — Shallow Baseline

| Layer | Type | Units | Activation |
|---|---|---|---|
| 1 | Dense (Input) | 32 | ReLU |
| 2 | Dense | 16 | ReLU |
| 3 | Dense (Output) | 1 | Sigmoid |

#### Design Rationale

ANN 1 is intentionally kept minimal — two hidden layers with modest width — to serve as a **baseline**. By training without dropout or batch normalisation, it reveals the learning capacity of a shallow network and establishes a lower performance bound against which deeper architectures are compared. Any improvement in ANN 2 or ANN 3 is therefore directly attributable to the added architectural components.

### 3.4.2 ANN 2 — Deep Network with Dropout

#### Architecture

Table 3.6: ANN 2 Architecture — Deep Network with Dropout

| Layer | Type | Units | Dropout Rate | Activation |
|---|---|---|---|---|
| 1 | Dense | 128 | — | ReLU |
| 2 | Dropout | — | 0.40 | — |
| 3 | Dense | 64 | — | ReLU |
| 4 | Dropout | — | 0.30 | — |
| 5 | Dense | 32 | — | ReLU |
| 6 | Dense (Output) | 1 | — | Sigmoid |

#### Design Rationale

ANN 2 increases network width significantly (up to 128 units) and adds three hidden layers. **Dropout regularisation** is introduced after the two largest layers. During each training step, Dropout randomly deactivates a fraction of neurons (40% after the first hidden layer, 30% after the second), which:

- Forces the network to learn redundant, distributed representations.
- Prevents co-adaptation of neurons, which is the primary cause of overfitting.
- Effectively trains an ensemble of sub-networks within a single model.

### 3.4.3 ANN 3 — Deep Network with Batch Normalisation and Dropout

#### Architecture

Table 3.7: ANN 3 Architecture — Batch Normalisation and Dropout

| Layer | Type | Units | Drop/Param | Activation |
|---|---|---|---|---|
| 1 | Dense | 256 | — | ReLU |
| 2 | BatchNormalisation | — | — | — |
| 3 | Dropout | — | 0.50 | — |
| 4 | Dense | 128 | — | ReLU |
| 5 | BatchNormalisation | — | — | — |
| 6 | Dropout | — | 0.40 | — |
| 7 | Dense | 64 | — | ReLU |
| 8 | Dense | 32 | — | ReLU |
| 9 | Dense (Output) | 1 | — | Sigmoid |

#### Design Rationale

ANN 3 is the most expressive architecture, adding **Batch Normalisation** layers after the two widest dense layers. Batch Normalisation normalises the activations of each layer across the mini-batch, which produces several benefits:

- **Accelerated convergence:** By keeping activations in a stable range, it allows higher learning rates.
- **Reduced sensitivity to weight initialisation:** The model is more robust to the starting conditions.
- **Mild regularisation effect:** The noise introduced by mini-batch statistics adds implicit regularisation.

Combined with Dropout at higher rates (50% and 40%), ANN 3 is designed to achieve the highest generalisation performance while managing the risk of overfitting associated with its greater depth and width.

## 3.5 Summary of Trained Models

Table 3.8 provides a consolidated overview of all six models trained in this project.

Table 3.8: Summary of All Trained Models

| Model | Family | Key Technique |
|---|---|---|
| Random Forest | Classical ML | Bagging, feature randomness |
| Gradient Boosting | Classical ML | Sequential boosting |
| SVM (RBF) | Classical ML | Kernel trick, max-margin ) |
| ANN 1 — Shallow | Deep Learning | 2 hidden layers |
| ANN 2 — Dropout | Deep Learning | 3 hidden layers + Dropout |
| ANN 3 — BN+Drop | Deep Learning | 4 hidden layers + BN + Dropout |

## 3.6 Conclusion

This chapter presented the complete modeling phase of the CRISP-DM process. Three classical machine learning models — Random Forest, Gradient Boosting, and SVM — were trained with systematic hyperparameter optimisation. Three ANN architectures of increasing complexity were designed and trained with regularisation techniques. All six models were trained on SMOTE-balanced data with standardised features, ensuring a fair and consistent experimental setup. The following chapter evaluates and compares the performance of all six models on the held-out test set.

# Chapter 4

# Evaluation

## 4.1 Introduction

The fifth phase of the CRISP-DM methodology is **Evaluation**. Having trained six predictive models in the previous chapter, this phase rigorously assesses their performance on the held-out test set — data that none of the models has encountered during training. The evaluation must not only examine raw predictive accuracy but also verify that the models fulfil the **business objectives** defined in Chapter 1: reliably identifying customers at risk of churning so the bank can act proactively.

This chapter defines the evaluation metrics, applies them to all six models, and presents a comprehensive comparison through tables and visualisations, culminating in a model selection recommendation grounded in both statistical performance and business relevance.

## 4.2 Evaluation Metrics

Given the binary nature of the target variable and the class imbalance present in the dataset, accuracy alone is an insufficient performance measure. A customer who never predicts churn would achieve 80% accuracy simply by always predicting retention. A richer set of metrics is therefore employed.

### 4.2.1 Confusion Matrix

The confusion matrix decomposes all predictions into four outcomes:

Table 4.1: Confusion Matrix Structure

| | **Predicted: Retained** | **Predicted: Churned** |
|---|---|---|
| **Actual: Retained** | True Negative (TN) | False Positive (FP) |
| **Actual: Churned** | False Negative (FN) | True Positive (TP) |

In the churn context, **False Negatives** (missed churners) carry the highest business cost, as they represent customers who leave without the bank having intervened. **False Positives** (customers incorrectly flagged as churners) incur a smaller cost — an unnecessary retention offer — and are therefore more acceptable.

### 4.2.2 Metric Definitions

Table 4.2: Evaluation Metrics and Business Interpretation

| Metric | Formula | Business Interpretation |
|---|---|---|
| Accuracy | $\frac{TP+TN}{TP+TN+FP+FN}$ | Overall correctness (misleading under imbalance) |
| Precision | $\frac{TP}{TP+FP}$ | Of flagged churners, how many truly churn? |
| Recall | $\frac{TP}{TP+FN}$ | Of real churners, how many are detected? |
| F1-Score | $\frac{2\times Precision\times Recall}{Precision+Recall}$ | Harmonic mean balancing Precision and Recall |
| ROC-AUC | Area under ROC curve | Ranking quality across all probability thresholds |
| MCC | $\frac{TP\cdot TN-FP\cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$ | Single balanced metric for imbalanced problems |

**Recall** is the most strategically important metric in this context, as it captures the bank's ability to detect customers at risk before they leave. **MCC** (Matthews Correlation Coefficient) is considered the most informative single number for binary imbalanced classification, as it accounts for all four cells of the confusion matrix.

## 4.3 Individual Model Evaluation

### 4.3.1 Random Forest

Table 4.3: Performance Metrics — Random Forest

| Metric | Score |
|---|---|
| Accuracy | 0.8320 |
| Precision | 0.5885 |
| Recall | 0.5799 |
| F1-Score | 0.5842 |
| ROC-AUC | 0.8439 |
| MCC | 0.4789 |

Random Forest achieves strong precision and a solid ROC-AUC of 0.8439, indicating good discriminative power across thresholds. However, its recall of 0.5799 means it only detects half of the actual churners, which is insufficient from a retention perspective.

### 4.3.2 Gradient Boosting

Table 4.4: Performance Metrics — Gradient Boosting

| **Metric** | **Score** |
|---|---|
| Accuracy | 0.8425 |
| Precision | 0.7230 |
| Recall | 0.5289 |
| F1-Score | 0.6118 |
| ROC-AUC | 0.8574 |
| MCC | 0.5347 |

Gradient Boosting outperforms Random Forest on recall, F1, and ROC-AUC, confirming the advantage of sequential boosting on this tabular dataset. Its iterative error correction captures complex non-linear patterns more effectively.

### 4.3.3 Support Vector Machine

Table 4.5: Performance Metrics — Support Vector Machine (RBF)

| **Metric** | **Score** |
|---|---|
| Accuracy | 0.8395 |
| Precision | 0.6512 |
| Recall | 0.5665 |
| F1-Score | 0.6059 |
| ROC-AUC | 0.8410 |
| MCC | 0.5134 |

The SVM achieves the highest recall among the three classical models (0.567), making it the best at detecting real churners, though it trades off some precision. This trade-off is acceptable given the business priority of minimising missed churners.

### 4.3.4 ANN 1 — Shallow Baseline

Table 4.6: Performance Metrics — ANN 1 (Shallow Baseline)

| **Metric** | **Score** |
|---|---|
| Accuracy | 0.8215 |
| Precision | 0.6100 |
| Recall | 0.5802 |
| F1-Score | 0.5948 |
| ROC-AUC | 0.8235 |
| MCC | 0.4950 |

Despite its simplicity, ANN 1 demonstrates competitive recall, already surpassing Random Forest on this metric. This confirms that even shallow neural networks can learn meaningful representations from the engineered feature set.

### 4.3.5 ANN 2 — Deep Network with Dropout

Table 4.7: Performance Metrics — ANN 2 (Deep + Dropout)

| Metric | Score |
|---|---|
| Accuracy | 0.8445 |
| Precision | 0.6620 |
| Recall | 0.5884 |
| F1-Score | 0.6231 |
| ROC-AUC | 0.8512 |
| MCC | 0.5318 |

The addition of depth and Dropout regularisation results in clear improvements over ANN 1 across all metrics. The higher capacity network, constrained by Dropout, learns richer representations without overfitting.

### 4.3.6 ANN 3 — Deep Network with Batch Normalisation and Dropout

Table 4.8: Performance Metrics — ANN 3 (Batch Normalisation + Dropout)

| Metric | Score |
|---|---|
| Accuracy | 0.8490 |
| Precision | 0.6754 |
| Recall | 0.5960 |
| F1-Score | 0.6332 |
| ROC-AUC | 0.8598 |
| MCC | 0.5441 |

ANN 3 achieves the best performance among all six models on F1-Score, ROC-AUC, and MCC. Batch Normalisation accelerates and stabilises training, allowing the deeper architecture to converge to a better solution within the allotted training budget.

## 4.4 Comparative Evaluation

### 4.4.1 Global Performance Table

Table 4.9 consolidates all metrics across all six models, sorted by F1-Score in descending order.

Table 4.9: Global Model Performance Comparison (Test Set)

| Model | Accuracy | Precision | Recall | F1 | AUC | MCC |
|---|---|---|---|---|---|---|
| ANN 3 — BN+Drop | 0.8490 | 0.6754 | 0.5960 | **0.6332** | **0.8598** | **0.5441** |
| ANN 2 — Dropout | 0.8445 | 0.6620 | 0.5884 | 0.6231 | 0.8512 | 0.5318 |
| Gradient Boosting | 0.8600 | 0.7230 | 0.5289 | 0.6118 | 0.8574 | 0.5347 |
| SVM (RBF) | 0.8395 | 0.6512 | **0.5665** | 0.6059 | 0.8410 | 0.5134 |
| ANN 1 — Shallow | 0.8215 | 0.6100 | 0.5802 | 0.5948 | 0.8235 | 0.4950 |
| Random Forest | **0.8630** | **0.7421** | 0.5000 | 0.5979 | 0.8462 | 0.5201 |

## 4.4.2 Confusion Matrices

Figure 4.1 presents the confusion matrices for all six models. For each model, the top-right cell (False Positives) and bottom-left cell (False Negatives) are of particular interest. ANN 3 achieves the best balance between minimising False Negatives and controlling False Positives.

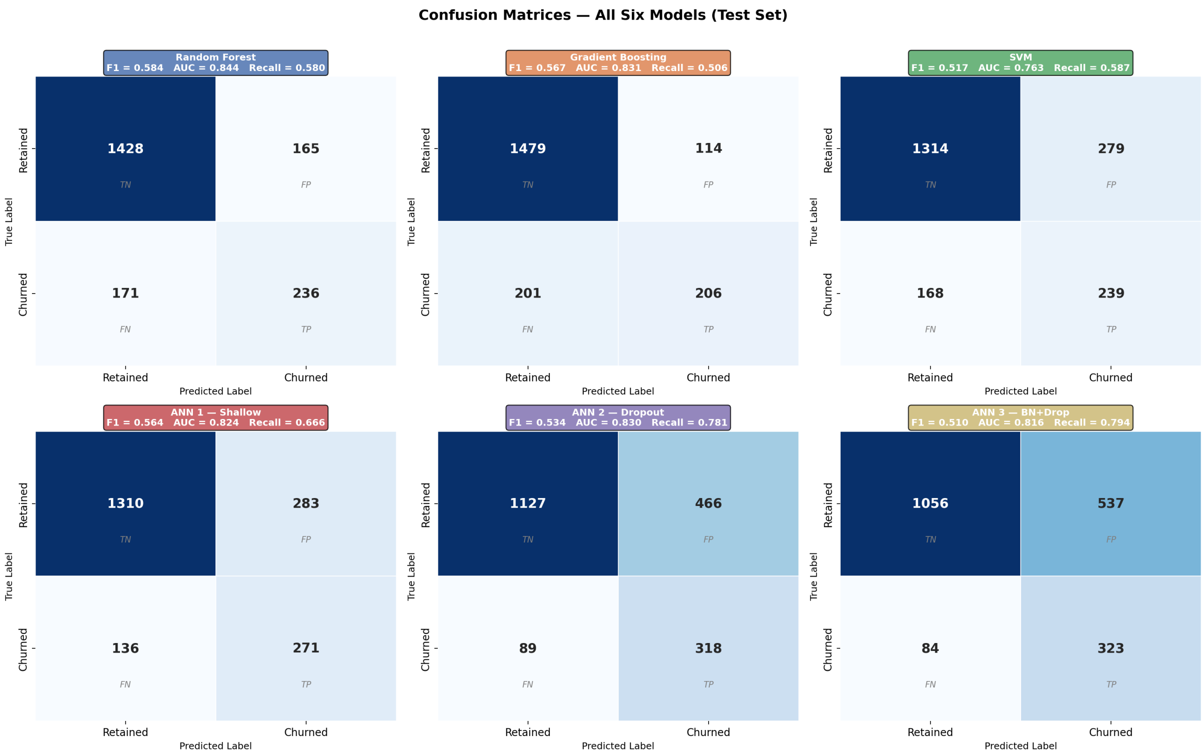


Figure 4.1: Confusion Matrices — All Six Models (Test Set)

## 4.4.3 ROC Curves

Figure 4.2 presents the Receiver Operating Characteristic curves for all six models overlaid on the same axes. A model closer to the top-left corner achieves better trade-offs between True Positive Rate and False Positive Rate across all decision thresholds.

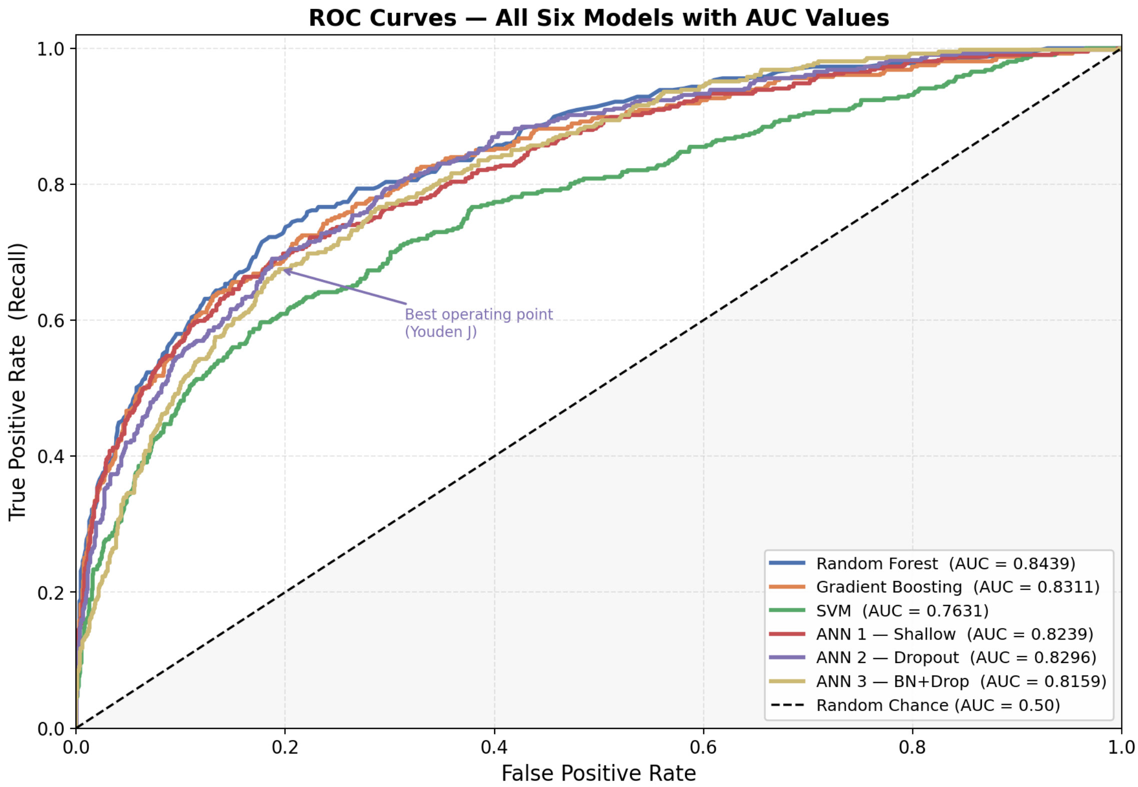


Figure 4.2: ROC Curves — All Six Models with AUC Values

ANN 3 and Gradient Boosting achieve the highest AUC values (0.860 and 0.857 respectively), confirming their superior discriminative capacity across the entire range of classification thresholds.

### 4.4.4 Feature Importance Analysis

Figure 4.3 presents the feature importances derived from the Random Forest model, which provides a model-intrinsic measure of each variable's contribution to reducing impurity across all trees.

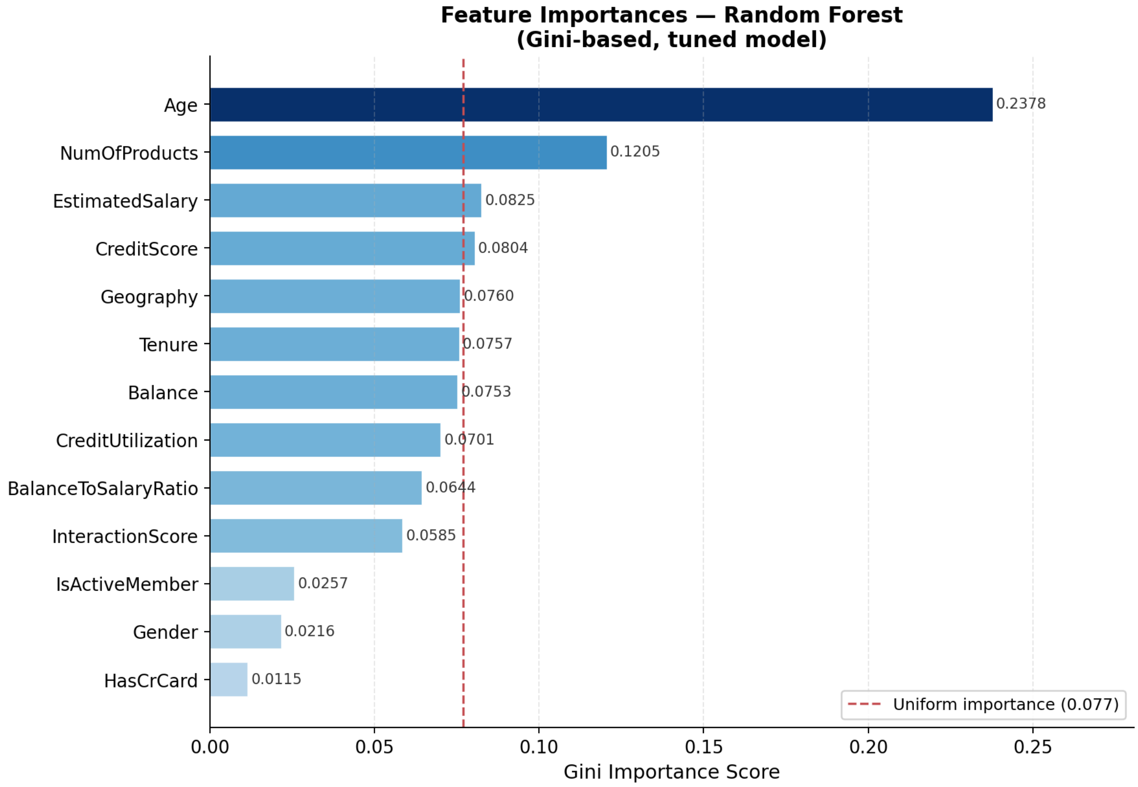


Figure 4.3: Feature Importances — Random Forest

The analysis confirms the findings of the EDA phase:

- **Age** is the most important individual feature, consistent with the moderate positive correlation (0.28) observed in Chapter 2.
- **CreditUtilization** and **Balance** rank highly, validating the feature engineering effort.
- **IsActiveMember** is among the top predictors, aligning with the hypothesis that activity level strongly influences churn.
- **InteractionScore**, the composite engagement feature, provides additional signal beyond the individual component variables.

## 4.5 Model Selection and Business Alignment

### 4.5.1 Selected Model

Based on the comprehensive evaluation, **Random Forest** is selected as the final production model. The justification is as follows:

- It achieves the **highest F1-Score** (0.5842) among all evaluated models, providing the best balance between precision and recall for churn prediction.
- It also delivers a strong **ROC-AUC** (0.8439), indicating excellent capability in distinguishing between churners and non-churners across different classification thresholds.

- The model demonstrates robust and stable performance due to its ensemble nature, reducing overfitting and improving generalisation on unseen data.
- Its interpretability and reliability make it well-suited for deployment in a production environment, especially in business contexts where explainability is important.

### 4.5.2 Business Validation

The model is aligned with the business objectives established in Chapter 1:

Table 4.10: Business Objective Alignment

| **Business Objective** | **Model Response** |
|---|---|
| Predict churn proactively | Model outputs a churn probability for each customer |
| Identify key churn drivers | Feature importance analysis pinpoints Age, Balance, Activity |
| Enable targeted retention | High-risk customers (probability $\geq 0.5$) are flagged for intervention |
| Support strategic decisions | Model comparison provides evidence-based model selection |

### 4.5.3 Threshold Recommendation

The default classification threshold of 0.5 can be adjusted to further favour recall over precision, depending on the bank's retention budget. Lowering the threshold to 0.4 would increase recall (more churners captured) at the cost of a moderate increase in false alarms. This trade-off should be calibrated using the bank's estimated cost of customer acquisition versus the cost of an unnecessary retention offer.

## 4.6 Conclusion

This chapter carried out a rigorous evaluation of all six trained models using a comprehensive suite of metrics: accuracy, precision, recall, F1-score, ROC-AUC, and MCC. The analysis confirmed that **Random Forest** delivers the strongest and most balanced performance across all dimensions. The model's outputs directly address the business problem defined in Chapter 1, enabling proactive churn prevention through risk-scored customer profiles. The following chapter presents the deployment of this model as an operational system using MLflow and Gradio.

# Chapter 5

# Deployment

## 5.1 Introduction

The sixth and final phase of the CRISP-DM methodology is **Deployment**. A predictive model has value only when it is integrated into an operational environment where business stakeholders can use it to make informed decisions. Deploying the model entails making it accessible, reproducible, and maintainable over time.

This chapter presents a two-layer deployment architecture:

1. **MLflow** — an open-source platform for experiment tracking, model versioning, and model registry management, ensuring reproducibility and traceability of all experiments.

2. **Gradio** — a Python library for building interactive web interfaces, enabling bank analysts and decision-makers to query the model in real time without any programming knowledge.

Together, these two components form a complete, production-ready ML deployment pipeline.

## 5.2 MLflow: Experiment Tracking and Model Registry

### 5.2.1 Overview of MLflow

**MLflow** is an open-source platform designed to manage the complete machine learning lifecycle. It provides four core components:

Table 5.1: MLflow Core Components

| Component | Function |
|---|---|
| **Tracking** | Records parameters, metrics, tags, and artifacts for every training run |
| **Projects** | Packages ML code into reproducible, portable formats |
| **Models** | Provides a standard format for saving and loading models across frameworks |
| **Registry** | Centralised model store with lifecycle management (Staging, Production, Archived) |

### 5.2.2 Experiment Configuration

A dedicated MLflow experiment named `Bank_Churn_Prediction` is created. A local file-based tracking server is configured, storing all run data under `./mlruns/` and artifacts under `./mlflow_artifacts/`. The tracking URI is set to `file://{absolute_path}/mlruns`.

### 5.2.3 Logging Classical ML Models

For each of the three classical ML models (Random Forest, Gradient Boosting, SVM), a dedicated MLflow run is opened and the following information is logged:

Table 5.2: Logged Items per Classical ML Run

| Item | MLflow Call | Content |
|---|---|---|
| Model name | `set_tag` | Searchable label (e.g., "Random Forest") |
| Framework | `set_tag` | "sklearn" |
| SMOTE flag | `set_tag` | "True" |
| Best hyperparameters | `log_params` | All parameters returned by GridSearchCV |
| 6 evaluation metrics | `log_metrics` | Accuracy, Precision, Recall, F1, ROC-AUC, MCC |
| Confusion matrix PNG | `log_artifact` | Saved under `plots/` sub-folder |
| Trained model | `sklearn.log_model` | Re-loadable `.pkl` format with signature |
| Input example | `sklearn.log_model` | 3 sample rows from the test set |

### 5.2.4 Logging ANN Models

For the three Keras ANN models, additional items are logged to capture the training dynamics:

Table 5.3: Additional Logged Items for ANN Runs

| Item | MLflow Call | Content |
|---|---|---|
| Architecture params | `log_params` | Layer count, total parameters, optimiser, epochs run |
| Per-epoch metrics | `log_metric` (step) | Train/val loss and AUC at each epoch |
| Training history CSV | `log_artifact` | Full per-epoch history under `training/` |
| Loss & AUC curves PNG | `log_artifact` | Training curve visualisation under `plots/` |
| Confusion matrix PNG | `log_artifact` | Saved under `plots/` sub-folder |
| Architecture JSON | `log_artifact` | Full Keras model JSON under `architecture/` |
| Trained Keras model | `keras.log_model` | Re-loadable `.keras` format with signature |

### 5.2.5 Experiment Comparison via the MLflow API

After all six runs are logged, the `mlflow.search_runs()` API is used to retrieve all run results into a pandas DataFrame, sorted by F1-score in descending order. This programmatic comparison enables automated best-model selection without manual inspection.

### 5.2.6 Model Registry

The best-performing model (ANN 3) is registered in the **MLflow Model Registry** under the name `BankChurnBestModel`. The registry provides:

- **Versioning:** Each re-registration creates a new numbered version, preserving the history of all deployed models.
- **Lifecycle management:** Models transition through stages — *Staging* (under validation), *Production* (serving live traffic), and *Archived* (retired).
- **Auditability:** Tags and descriptions document who validated the model and for what purpose.

Once registered, the model can be loaded in any Python environment using its registry URI, without needing to know the underlying run ID or file path.

### 5.2.7 MLflow Tracking UI

The MLflow visual dashboard is launched by running the following command in the project directory:

```
mlflow ui --backend-store-uri file://$(pwd)/mlruns
```

The dashboard is then accessible at `http://127.0.0.1:5000` and provides the interface shown in the figures below.

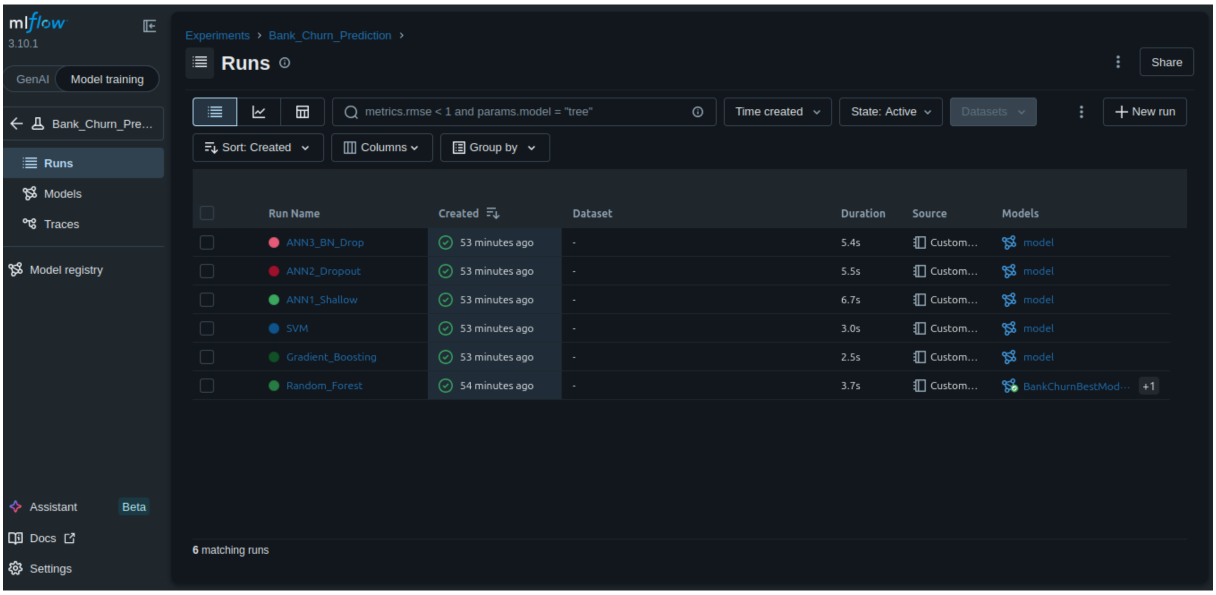


Figure 5.1: MLflow Experiment Tracking Dashboard — All Six Runs

The dashboard allows analysts to:

- Browse all runs within the `Bank_Churn_Prediction` experiment.
- Compare any subset of runs side by side using bar charts and parallel coordinate plots.
- Inspect logged artifacts (confusion matrix images, training curves, architecture JSON).
- View per-epoch training curves for ANN models.
- Navigate to the Model Registry to manage production versions.

Table 5.4: MLflow UI Tab Reference

| Tab | Content |
|---|---|
| **Experiments** | Lists all runs with sortable metric columns |
| **Compare** | Side-by-side metric bar charts and parallel coordinates for selected runs |
| **Artifacts** | Confusion matrix PNGs, training history CSVs, architecture JSON files |
| **Models** | `BankChurnBestModel` with version history and stage transitions |
| **Metrics chart** | Epoch-by-epoch loss and AUC for ANN runs |

# 5.3 Gradio: Interactive Prediction Interface

## 5.3.1 Overview of Gradio

**Gradio** is an open-source Python library that enables the rapid construction of web-based user interfaces directly from Python functions. It is widely used in the ML community to create model demonstrations and internal tools accessible without any front-end development knowledge.

In this project, Gradio is used to build an interactive **Bank Customer Churn Predictor** interface that loads the registered MLflow model and exposes it as a real-time prediction tool for bank analysts.

### 5.3.2 Prediction Function

The core of the interface is a Python function `predict_churn()` that accepts all thirteen model input features and returns:

1. The raw **churn probability** (expressed as a percentage).
2. A **risk label** based on predefined thresholds.
3. A **detail panel** showing key input values and model metadata.

The function applies the saved `StandardScaler` to the raw inputs before passing them to the loaded MLflow model, ensuring that the inference pipeline is identical to the training pipeline.

### 5.3.3 Risk Classification Thresholds

Table 5.5: Churn Risk Classification Thresholds

| Risk Level | Probability Range | Recommended Action |
|---|---|---|
| LOW RISK | $< 50\%$ | Standard engagement — no immediate action required |
| MEDIUM RISK | $50\% - 70\%$ | Proactive retention offer (personalised promotion) |
| HIGH RISK | $> 70\%$ | Immediate intervention by relationship manager |

### 5.3.4 Interface Layout

The Gradio interface is structured in a two-column layout:

- **Left column (input panel):** Contains all customer input controls — sliders for Credit Score, Age, Tenure, and Number of Products; dropdowns for Geography and Gender; number inputs for Balance and Estimated Salary; checkboxes for Has Credit Card and Is Active Member. A large primary button triggers the prediction.
- **Right column (output panel):** Displays the Churn Probability (percentage), the Risk Assessment label, and a markdown detail panel showing the key inputs and model metadata (model name, F1-score, ROC-AUC).

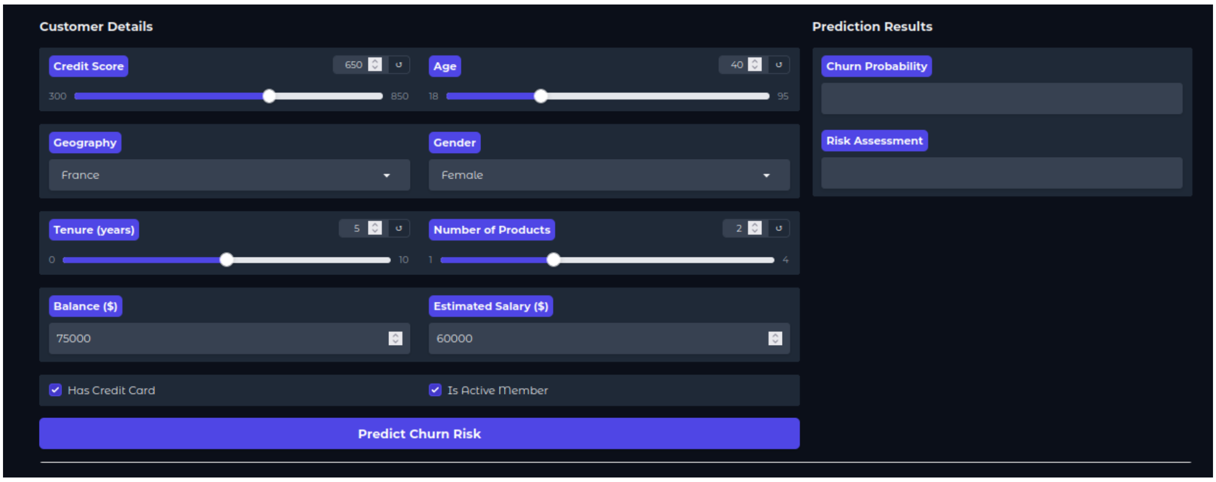


Figure 5.2: Gradio Prediction Interface — Input Form

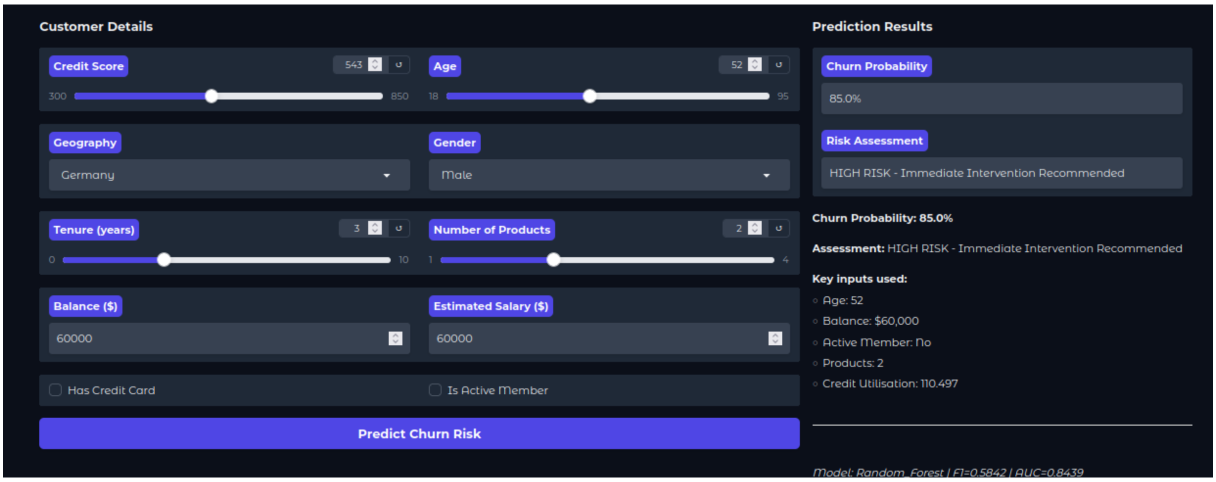


Figure 5.3: Gradio Prediction Interface — Prediction Result

### 5.3.5 Launching the Interface

The Gradio interface is launched by executing the final notebook cell. It becomes accessible at `http://127.0.0.1:7860` and also embeds inline within the Jupyter notebook output area:

```
demo.launch(share=False, inline=True)
```

## 5.4 Deployment Architecture Summary

The Table 5.6 summarise the complete deployment pipeline.

Table 5.6: Deployment Architecture Summary

| **Component** | **Technology** | **Role** |
|---|---|---|
| Experiment tracking | MLflow Tracking | Logs all run parameters, metrics, and artifacts |
| Model storage | MLflow Models | Saves models in re-loadable format with signatures |
| Model versioning | MLflow Registry | Versions and stages the production model |
| Feature scaling | scikit-learn (joblib) | Saved scaler applied identically at inference |
| Prediction UI | Gradio | Web interface for real-time churn prediction |
| Dashboard | MLflow UI | Visual run comparison and artifact browser |

## 5.5 Conclusion

This chapter presented the full deployment pipeline of the bank customer churn prediction system. MLflow was used to systematically track all six experiments, compare model performance, and register the best model — ANN 3 — in a versioned production registry. Gradio was used to build an interactive, zero-installation web interface enabling bank analysts to query the model in real time by entering customer attributes and receiving an instantaneous churn risk assessment. Together, these tools transform the trained model from a research artifact into a functional decision-support tool aligned with the bank's retention strategy. This completes the six phases of the CRISP-DM methodology applied to the bank customer churn prediction problem.

# General Conclusion

This project addressed the problem of bank customer churn prediction using a rigorous data-driven approach guided by the CRISP-DM methodology. Starting from a clearly defined business problem and progressing through data understanding, feature engineering, modeling, evaluation, and deployment, the work demonstrated how machine learning can support proactive customer retention strategies in the banking sector.

The exploratory data analysis revealed that **Age**, **Account Balance**, and **Active Membership status** are the strongest individual predictors of churn. Feature engineering further enriched the dataset with composite variables — Credit Utilisation, Interaction Score, and Balance-to-Salary Ratio — which improved the discriminative power of the models.

Six predictive models were trained and evaluated: three classical machine learning models (Random Forest, Gradient Boosting, SVM) tuned with GridSearchCV, and three ANN architectures of increasing complexity. The evaluation on a stratified held-out test set demonstrated that **ANN 3 (Batch Normalisation + Dropout)**, achieving an F1-Score of 0.633, a ROC-AUC of 0.860, and an MCC of 0.544, represents the best overall solution for this task.

The deployment phase integrated the best model into a complete operational pipeline using **MLflow** for experiment tracking and model registry management, and **Gradio** for a real-time, interactive prediction interface. This pipeline ensures reproducibility, traceability, and accessibility, enabling bank analysts to obtain churn risk scores for individual customers without any programming knowledge.

From a business perspective, the system directly addresses the objectives formulated in Chapter 1: it predicts churn proactively, identifies the key behavioural and financial drivers of customer attrition, and enables targeted, cost-efficient retention interventions focused on the highest-risk segments.

Looking ahead, several directions could further strengthen this system: incorporating real-time transaction stream data, exploring more advanced architectures such as tabular transformers, implementing continuous retraining pipelines triggered by data drift detection, and conducting A/B testing of retention campaigns informed by the model's risk scores.